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Leveraging Recycled Rubber and Organic Plugging Materials for Loss Circulation Pill

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Abstract

This study addresses lost circulation of drilling fluids as one of the most challenging drilling issues. The study proved the superiority of unique composition of blend of recycled rubber material with organic plugging materials to enhance fluid loss control. The success criteria have been set as the following: (1) Cost efficient compared to available solutions (2) Superior performance to conventional LCMs (3) Supply sustainability and environmental impact. A comparative laboratory assessment is conducted on conventional as wells as the newly developed LCM pills. These pills were evaluated using a 20/40 mesh sand bed at different sand bed compaction levels. The LCM pill is placed on the top of the sand bed and mounted in a modified API test cell that is connected to a nitrogen gas cylinder to apply required pressure on top of the loss control slurry. The applied pressure simulated overbalance pressure during the treatment of a loss zone. Experimental results have shown that conventional LCM pill escaped through the permeable sand bed matrix immediately after the application of 100 psi overbalance pressure. In the developed LCM case, though, there was a slight drop in the slurry level due to an initial slurry invasion into the sand bed matrix, but no further drop has been noticed afterward. Also, conventional LCM pill formed thicker cake compared to the developed pills. This is because thinner cake reduces the likelihood of pipe sticking issues. The visual and digital data obtained clearly have shown excellent sealing and blocking of 2 mm slotted disc at 500 psi overbalance pressure both at room temperature and at 250 F temperature as it quickly created a bridging effect. The chemicals requirements needed are evaluated and compared to conventional LCM pill where supply of the developed LCM pill can be sourced locally with reduced cost. In addition, the recycled rubber contributes to a sustainable circular economy. This study contributed to circular economy by providing enhanced solutions to cure loss circulation. The engineered LCM pill is superior to conventional LCM pills in terms of performance in sealing and blocking along with cost effectiveness in loss circulation zones.

Introduction

Drilling fluids are critical to maintain wellbore integrity and enhance operational efficiency. Chemical composition, fluids density, and rheological properties are rated among top priorities in designing and optimizing drilling fluids as they provide tailored cost-efficient performance. The key drilling fluids functions include carrying cuttings, downhole pressure stability, filter cake formation, and overcoming

challenging subsurface environment [1] [2]. To ensure effective and efficient performance, drilling fluids must maintain cuttings suspension to transport them to the surface without settling, which otherwise contributes to increased torque and drag during drilling operations. The efficiency of this suspension mechanism is primarily governed by the fluid's density, rheological properties, and circulation velocity [3]. Besides, solids presence in drilling fluids plays a critical role by filling pore spaces within the formation and forming a mechanically protective layer known as a filter cake. Beyond reinforcing the structural integrity of the wellbore wall, the filter cake assists in reducing fluid invasion into permeable formations, reduce associated environmental risks, and facilitate improved cement placement by providing a stable interface [4]. These functions collectively contribute to wellbore stability and drilling performance, but their effectiveness largely relies on using properly engineered fluids with additives tailored to operational demands and subsurface conditions [5]. However, despite careful fluid design, unanticipated formation heterogeneities and pressure differentials can still lead to fluid losses into the formation. Such loss of circulation events represents one of the most persistent and costly challenges in drilling operations, necessitating specialized fluid systems and mitigation strategies.

Lost circulation is a significant challenge that frequently arises during drilling operations. It occurs when drilling mud seeps into the formation either partially or entirely, which is an unintended and undesirable situation. This issue can lead to several serious problems, such as stuck pipes, damage to the formation, blowouts, and instability in the wellbore. These problems often result in delays and higher expenses for the operation. As a result, it's standard practice in the industry to minimize these incidents as much as possible. The probability of lost circulation can be influenced by a variety of factors, including the type and weight of the drilling fluid, its rheological properties, the pressure from the pumps, and the pore pressure of the formation. Lost circulation can be categorized into different types based on the severity and nature of the fluid loss. The primary types of lost circulation are seepage, partial and total losses. Seepage loss occurs when the rate of fluid loss is in the range of 1 to 10 barrels per hour. It can happen in any type of formation but more likely when drilling through large porous, unconsolidated or highly permeable formation such as loose sand and gravels. This type is usually caused when the bridging material is not large enough to form a seal. Partial losses are generally more critical than seepage losses and typically require the addition of lost-circulation material (LCM) to reduce the losses. This type of loss can occur in small natural fractures and barely opened induced fractures, and the severity of the loss is in the range of 10 to 500 barrels per hour. The most extreme form of loss circulation is the severe or total loss where the entire drilling fluid is lost into the formation. It can occur in long and open gravel, large natural horizontal fractures, and caverns [6] [7] [8].

Loss circulation management continues to be a critical focus in both upstream oil and gas operations and academic research. Studying prevention and remedial methods requires a detailed understanding of the formation types responsible for fluid losses and the mechanisms governing them. A thorough evaluation of downhole conditions, including fracture geometry, pore pressure, and lithology needed to select the appropriate lost circulation material (LCM) [9], [10], [11] Click or tap here to enter text. Natural polymers such as starch and carboxymethyl cellulose (CMC) are widely used in drilling fluids to minimize fluid invasion and enhance filter cake quality. Anderson and Edwards [12] demonstrated the effectiveness of polymer-based additives in potassium mud systems for stabilizing reactive shale formations. Conventional additives are often ineffective in ultra-low permeability formations due to their inability to penetrate nano-scale pore throats. When shale interacts with water-based drilling fluids, hydration and swelling can occur, leading to severe wellbore instability. Nanoparticle additives such as clay/SiO₂ and TiO₂ have shown promise in mitigating these effects. Their small size (1–500 nm) enables deeper penetration into nanopores, enhancing sealing efficiency and improving the rheological and filtration properties of the fluid [13].

LCMs are solid or semi-solid materials designed to mitigate fluid losses by physically bridging, sealing, or plugging loss zones. Their performance is influenced by factors such as particle size distribution, mechanical strength, and compatibility with drilling fluids. Industry standards classify LCMs into seven categories: flaky, granular, fibrous, blended, water/acid-soluble, dehydratable, and nanoparticle-based materials. Each

category exhibits distinct physical and chemical properties suited for specific loss mechanisms and formation types [14]. There are still significant knowledge gaps in addressing fluid losses encountered while drilling shallow sections of both onshore and offshore oil and gas wells. These sections are commonly composed of granular materials such as sand, gravel, rubble, and interspersed silt or clay, often feature well-defined permeable channels with hydraulic conductivities reaching up to 1000 Darcy. Such geological characteristics frequently result in moderate to severe mud losses during both conventional and unconventional drilling operations, especially under conditions of poor compaction and weak formation bonding. Also, partial to total losses may arise from induced fractures in formations exhibiting low fracture gradients, narrow mud weight windows, or pressure imbalance. The size and geometry of fractures can range from microns to several centimeters, governed by the degree of consolidation, chemical alteration, and the magnitude of overbalance pressure applied during drilling. This study attempts to address the limitation of conventional LCM pills and elevate its performance with an experimental approach. The study compares the performance of two industry standard LCM pills and two newly engineered LCM formulations.

Materials and Methods

Preparation for Loss Circulation Simulation

The sand bed is placed in a see-through test cell that can withstand pressures more than 100 psi. The sand bed represents unconsolidated or poorly consolidated formations such as sandstone or loose gravel zones that are high permeability. Loss of drilling fluids occur in similar environments. A circular metallic screen installed in the cell at the base to retain the sand while allowing fluid flow. This screen has a solid circular border at the periphery and a 60-mesh (250 microns) central screen positioned directly above the 10 mm exit port in the bottom lid of the assembly, see [Figure 1](#). The sand used in the test was 20/40 mesh sand which consists of particles ranging in size from 425 to 850 microns. This size range is larger than the 60 mesh screen used at the bottom of the test cell to ensure particle retention during testing. The sand bed dimensions matched the inner diameter of the test cell, with a height of 105 mm and a diameter of 76 mm. Two types of sand beds were prepared to simulate distinct loss circulation scenarios: a well-compacted sand bed representing a partial loss zone, and a poorly compacted sand bed simulating a total loss zone. A well-compacted but unconsolidated sand bed may exhibit permeability values up to 500 Darcy, while a poorly compacted bed may reach permeability levels of up to 1000 Darcy, depending on the degree of compaction. These conditions mimic subsurface loss zones with varying degrees of circulation loss potential. To simulate realistic downhole saturation conditions, the sand was pre-wetted and fully saturated with water before placement. Once the required volume of sand was added to reach a pre-marked level inside the test cell, it was compacted and leveled using a foot-operated stand equipped with a flat circular disc at its base, see [Figure 2](#). An axial force was applied by pressing downward on the stand to achieve uniform compaction. [Figure 2](#) presents the resulting prepared sand bed. A new sand bed of identical dimensions was constructed for each individual test, ensuring consistency by applying a similar compaction force. A higher compaction force was used when simulating partial loss zones.



Figure 1—Test cell filled partially with 20/40 sand.

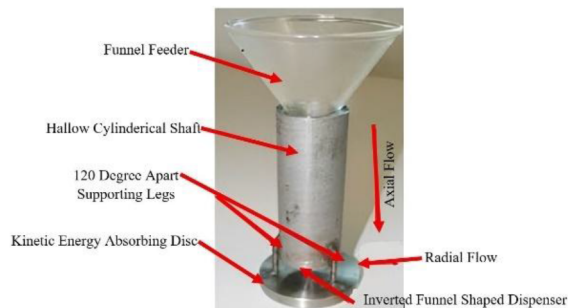


Figure 2—The LCM placement device throughout the study.

Energy-Based Design and Function of the LCM Slurry Placement Apparatus

A novel apparatus is developed to simulate the placement of lost circulation material (LCM) slurry. It is driven from the spatial and velocity dependent behavior of potential and kinetic energies in free falling fluids. This system, as seen in [fig. 2](#), is engineered to ensure controlled deposition of LCM slurry atop a preconditioned sand bed, to reduce the effect of mechanical disturbance and to maintain the structural integrity of the test substrate. The device consists of a vertical flow path that allows gravitational acceleration of the slurry followed by a controlled reduction in both potential and kinetic energy as the fluid approaches the energy absorbing interface located above the sand bed. Upon contact, the energy absorbing disc dissipates axial momentum and induces a radial flow pattern, effectively distributing the slurry laterally while minimizing localized impact forces. This redirection not only enhances hydrodynamic energy dissipation but also prevents erosion, compaction, or redistribution of the underlying particulate structure. The reduction in dynamic energy prior to slurry-sand contact is critical to maintaining the fidelity of the experimental test bed. By eliminating mechanical degradation during slurry placement, the apparatus enables reproducible and accurate performance evaluations of various LCM systems under controlled, undisturbed conditions. The method ensures that measured sealing efficiency, bridging behavior, and filter cake characteristics are attributable solely to the LCM formulation, not to inconsistencies in substrate integrity.

Carrier Fluid Selection and Slurry Preparation

The carrier fluid for the LCM pill formulation can be any water or saltwater but it should be viscosified with additives to suspend and transport the particulate components effectively. Operational practice typically involves utilizing 50 to 100 barrels of the active drilling fluid, commonly the filling wellbore, as the carrier medium for LCM deployment in losses zone. In this study, a bentonite-based mud system, for drilling the surface sections, was selected as the carrier fluid. The LCM components were incorporated into this base fluid using a high shear, high speed mixing unit to ensure uniform dispersion and to achieve a homogenous loss control slurry. The rheological and compositional characteristics of this slurry were tailored to facilitate both suspension stability and effective placement into the loss zone. The specific formulation of the

bentonite mud used as the carrier fluid in this study is presented in [Table 1](#). This formulation was utilized to evaluate the performance of various LCM slurries under simulated conditions of partial and total circulation loss.

Table 1—Bentonite LCM Pill Design Formulation

Mud Additives	Field Quantity	Lab Quantity
Fresh Water	0.97 bbl	339.8 (cc)
Soda Ash (Na ₂ CO ₃)	0.5 ppb	0.5 gm
Bentonite	25 ppb	25 gm
Caustic Soda (NaOH)	0.5 ppb	0.5 gm

LCM Pills

Four LCM pill formulations were evaluated to assess their effectiveness to cure loss circulation of different severities: two industry standard formulations, and two engineered formulations incorporating organic materials and recycled rubber. The conventional LCM pills used for partial and total loss scenarios are provided in [Tables 2](#) and [3](#), respectively. These formulations reflect standard industry practice and are routinely deployed in both conventional and unconventional drilling operations to address loss events. [Tables 4](#) and [5](#) present the formulations of the newly developed LCM pills. These novel systems incorporate synthesized materials that serve as LCM additives. These substitute additives aim to reduce operational costs while maintaining or improving performance in loss control applications. The study compares the performance of organic materials relative to conventional formulations in terms of cost effectiveness along with performance LCM solutions tailored to operational needs.

Table 2—Conventional LCM Pill Designed for Partial Loss Control

LCM Additives	Median Particle Size-D50 (microns)	Concentration (ppb)/bbl
WalNut Shells	1200-1500	15
Sure Seal	400-600	10
Bicarbonate	1200-1400	10
Marble C	600	15
Marble M	150	20
Bicarbonate	50	10
Corn Fibers	850	15
Sunsweep	-	0.2
Total Pill Concentration		95.2
Carrier Fluid		Bentonite Mud
LCM Products Used		63% Imported
		37% Local

Table 3—Conventional LCM Pill Designed for Total Loss Control

LCM Additives	Median Particle Size-D50 (microns)	Concentration (ppb)/bbl
WalNut Shells	1200-1500	15
Sure Seal	400-600	10
Barofiber C	1200-1400	10
Marble C	600	20
Marble M	150	20
Bicarbonate	50	10
Graphite	300-400	10
Corn Fibers	850	15
Bicarbonate	60	10
Sunsweep	-	0.2
Total Pill Concentration		120.2
Carrier Fluid		Bentonite Mud
LCM Products Used		67% Imported
		33% Local

Table 4—Engineered LCM Pill, Sealer A, for Partial Loss Control

LCM Additives	Median Particle Size-D50 (microns)	Concentration (ppb)/bbl
Date Seeds	1000-1200	15
Sure Seal	400-600	10
Date Tree Fibers	800-1000	10
Marble C	600	15
Marble M	150	20
Bicarbonate	50	10
Corn Fibers	850	15
Sunsweep	-	0.2
Total Pill Concentration		95.2
Carrier Fluid		Bentonite Mud
LCM Products Used		37% Imported
		63% Local

Table 5—Engineered LCM Pill, Sealer B, for Total Loss Control

LCM Additives	Median Particle Size-D50 (microns)	Concentration (ppb)/bbl
Date Seeds	1000-1200	15
Sure Seal	400-600	10
Date Tree Fibers	800-1000	10
Marble C	600	20
Marble M	150	20
Bicarbonate	50	10
Resin	-	10
Corn Fibers	850	15
Date Tree Fibers	-	10
Sunsweep	-	0.2
Total Pill Concentration		120.2
Carrier Fluid		Bentonite Mud
LCM Products Used		38% Imported
		62% Local

Experimental Procedure

Following the preparation of the 20/40 sand bed as previously described, a laboratory-scale volume (350 cc) of the loss control slurry was placed onto the sand surface using the designed LCM placement device to ensure uniform deposition without disturbing the integrity of the bed. This method effectively eliminated mechanical degradation of the test substrate during slurry placement. Then, the test assembly was mounted within a modified API filtration cell [figure 3](#), which was configured to accommodate a nitrogen gas line for applying controlled overbalance pressure onto the slurry. This setup simulates the pressure differential

encountered across a loss zone during actual field treatment conditions. Prior to pressurization, a 2–3 minute observation period was conducted under static conditions (i.e., with no externally applied pressure, only the slurry's hydrostatic head) to detect any immediate leakage or infiltration. All LCM slurry formulations demonstrated full containment, with no observable leakage or loss during this baseline phase. After the initial static observation, a constant overbalance pressure of 100 psi was applied to the slurry column. Visual monitoring, along with photographic and video documentation, was carried out throughout the test duration. Key performance indicators like slurry level reduction, depth of fluid invasion into the sand matrix, and filter cake thickness were recorded systematically. Each test was repeated multiple times to verify reproducibility and ensure statistical reliability of the observed results.

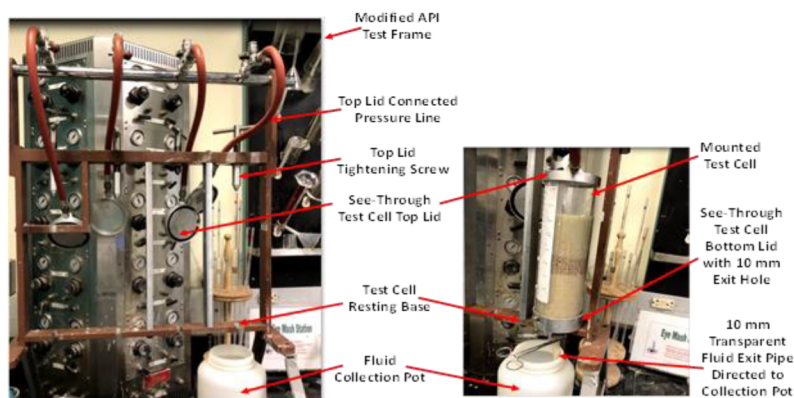


Figure 3—Modified Test Frame Used to Mount the See-Through Test Cell

Results and Discussion

The experimental results involve two loss circulation scenarios, partial and total losses, with each scenario investigated using both the conventional LCM pill and the newly developed enhanced LCM formulation introduced in this study.

Sealing Efficiency and Invasion Behavior in Partial Loss Conditions

The performance evaluation conducted under controlled laboratory conditions using a custom-designed test apparatus operating at a constant overbalance pressure of 100 psi. The conventional LCM pill showed limited sealing capabilities, as evidenced by initial visual inspections that showed a clean slurry interface with the high-permeability sand bed and no evidence of cake formation or fluid obstruction prior to pressurization. Upon applying pressure, the conventional pill failed to slow down fluid migration into the porous matrix leading to a complete and rapid decline in slurry level, see [fig. 4](#). This was accompanied by the appearance of cloudy carrier fluid in the outflow, indicating substantial slurry invasion and significant carrier fluid filtration. Post-test evaluation exposed the deposition of a thick filter cake ranging from 55 to 65 mm on the surface of the sand bed, see [table 6](#). This indicates inefficient filtration control and raises significant operational concerns related to thick cake formation. This deposition indicates poor solid bridging performance and increases the risk of differential sticking which can cause stuck pipe. As a result, drilling costs may rise, and future drilling and completion operations may become more complicated. Besides, the invasion depth extended beyond the entire length of the sand bed showing its failure to form an effective barrier in high-permeability formations. Deep invasion of solids can damage the formation, reduce near-wellbore stability, and negatively affect well productivity.

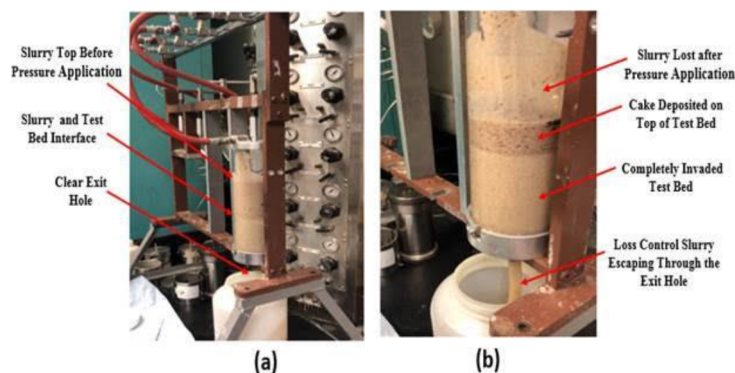


Figure 4—Photos of conventional LCM pill for partial losses scenario: (a) Before applying pressure; (b) After applying pressure.

Table 6—Conventional LCM pill for partial loss control under 100 psi pressure.

Measured Parameters	20/40 Sand Bed
Spurt Loss (cc)	All 350 cc Slurry Lost
Mud loss (cc)	
Total Leaf Off (cc)	
Seal/Plug Thickness	55-65 mm
Slurry Level Drop	Total
Invasion Depth	Beyond the Sand Bed Height (105 mm)

The enhanced LCM pill demonstrated significantly superior sealing performance under identical conditions. Before applying, similar initial slurry-bed interface observed without evidence of premature cake development yet its response to applied pressure differed fundamentally. After applying pressure, only a slight initial slurry drop was observed followed by rapid and complete cessation of slurry movement through the porous matrix indicating immediate sealing of the loss zone. The clear drainpipe after the test confirmed that very little carrier fluid had invaded showcasing strong sealing and fluid retention, see [fig.5](#). The measurements revealed the formation of a much thinner filter cake, ranging from 20 to 25 mm, which provides sufficient sealing at the loss interface without contributing excessive solids buildup that could risk mechanical pipe entrapment or impair future wellbore operations. Moreover, the enhanced pill restricted invasion depth to approximately 50 mm which is less than half the length of the 105 mm sand bed. This demonstrates better sealing behavior and bridging properties that effectively isolated the loss zone and prevented further fluid migration, see [table 7](#). The controlled cake thickness and limited invasion which can reduce formation damage potential and help for smooth cementing and completion processes. The enhanced sealing efficiency can be attributed to its optimized particle size distribution, tailored bridging agents, and engineered rheological properties that enable rapid pore-throat bridging and stable suspension behavior under dynamic drilling conditions.

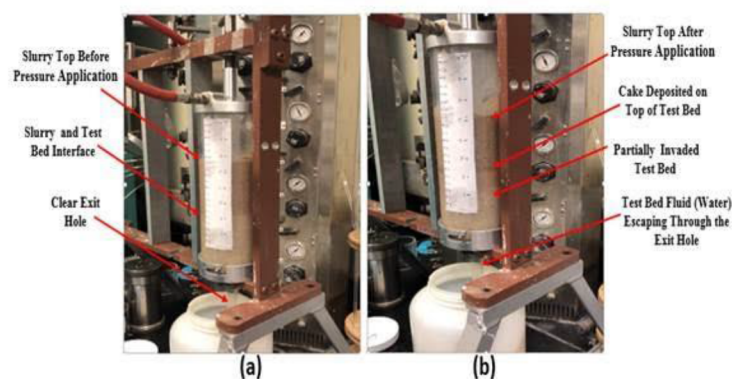


Figure 5—Photos of enhanced LCM pill for partial losses scenario: (a) Before applying pressure; (b) After applying pressure.

Table 7—Enhanced LCM pill for partial loss control under 100 psi pressure.

Measured Parameters	20/40 Sand Bed
Spurt Loss (cc)	0
Mud loss (cc)	
Total Leaf Off (cc)	
Seal/Plug Thickness	20-25 mm
Slurry Level Drop	40-45 mm
Invasion Depth	50 m

Sealing Efficiency and Invasion Behavior in Total Loss Conditions

The results also confirmed the differences in sealing performance, invasion depth, and cake deposition characteristics between the two LCM pills under total losses conditions. For the conventional LCM pill, initial visual inspection, before applying 100 psi pressure, showed a stable slurry interface with the high-permeability sand bed. There is no evidence of cake formation and a clear exit path in the transparent outlet pipe. After pressure application, a slight reduction in slurry level was observed followed by the complete cessation of slurry flow through the sand bed matrix, see [fig. 6](#). This behavior suggests that the conventional pill was able to initiate sealing after limited initial fluid invasion allowing the formation of a filter cake under the remaining slurry column and above the sand bed surface. The transparent outlet pipe remained clear throughout the test showing no significant fluid or slurry breakthrough occurred. While this demonstrates the conventional pill capacity to stop total loss, still the data revealed that this sealing was achieved after partial invasion into the sand bed as the invasion depth measured between 45 and 50 mm corresponding to approximately 50% of the sand bed height. These data are shown in [table 8](#) where it also observed that filter cake thickness was 15 to 20 mm that deposited at the slurry-sand interface. The slurry level dropped by 25 mm during the test.

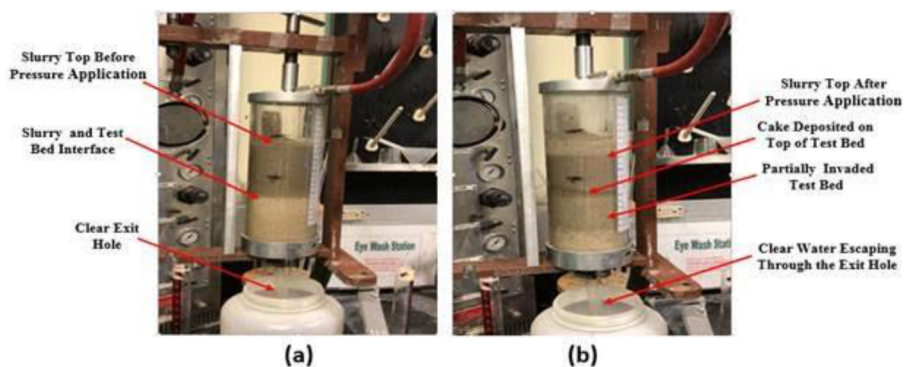


Figure 6—Photos of conventional LCM pill for total losses scenario: (a) Before applying pressure; (b) After applying pressure.

Table 8—Conventional LCM pill for total loss control under 100 psi pressure.

Measured Parameters	20/40 Sand Bed
Spurt Loss (cc)	0
Mud loss (cc)	
Total Leaf Off (cc)	
Seal/Plug Thickness	15-20 mm
Slurry Level Drop	25 mm
Invasion Depth	45-50 mm

The enhanced LCM pill demonstrated better performance under identical test conditions. Before pressurization, the slurry-sand interface was stable without premature cake formation. However, after applying 100 psi pressure, there was minor slurry level loss followed by effective sealing of the highly permeable sand bed. The outlet pipe remained entirely clear throughout the test that showed minimal carrier fluid migration, see [fig. 7](#). [Table 9](#) shows a high sealing efficiency of the enhanced pill. The measured invasion depth was limited to approximately 20 mm, significantly lower than that of the conventional pill, representing less than 20% of the sand bed height. Additionally, the filter cake formed was thinner, ranging from 6 to 8 mm, thus minimizing the risk of excessive solids accumulation on the borehole wall. The slurry level drop was also smaller compared to that recorded for the conventional formulation, indicating more immediate sealing response with less fluid loss prior to barrier establishment.

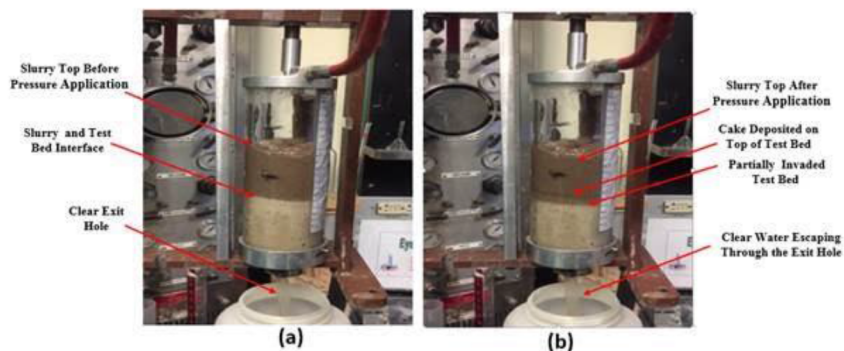


Figure 7—Photos of enhanced LCM pill for total losses scenario: (a) Before applying pressure; (b) After applying pressure.

Table 9—Conventional LCM pill for total loss control under 100 psi pressure.

Measured Parameters	20/40 Sand Bed
Spurt Loss (cc)	0
Mud loss (cc)	
Total Leaf Off (cc)	
Seal/Plug Thickness	6-8 mm
Slurry Level Drop	12 mm
Invasion Depth	20 mm

Conclusions

The enhanced LCM pills consistently showed better performance compared to conventional formulations in both partial and total loss circulation scenarios. Laboratory evaluations, including sand bed simulation tests, confirmed the enhanced pills' ability to rapidly establish effective seals in unconsolidated and high-permeability formations while minimizing fluid invasion, filter cake thickness, and formation damage. This improved sealing efficiency reduces differential sticking risks, enhances drill string mobility, and simplifies subsequent well construction operations. That, the lower fluid loss prior to sealing translates into reduced LCM consumption and fewer remediation cycles, offering clear operational and economic advantages. The demonstrated versatility and reliability of the enhanced pills across a range of loss conditions highlight their potential to enhance wellbore stability, safeguard reservoir integrity, and optimize overall drilling performance in complex geological environments.

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